Newton

a) $y = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ is LP-feasible. Every row $\frac{1}{2} + \frac{1}{2} = 1 \geq 1$ tight on all three and $0 \leq \frac{1}{2} \leq 1$ feasible

Cost $= 5(1/2) + 5(1/2) + 6(1/2) = 2.5 + 2.5 + 3 = 8$

b) Rounding. Round down to (0,0,0): fails every row (see H1's table) infeasible

Round up to (1, 1, 1): feasible (every row ≥ 1), cost = 16

c) z^* and LP gap. From H1, $z^* = 10$. Is $y = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ actually the LP optimum, i.e. is $z_{LP} = 8$? Yes

Solving the LP with any solver (src/lp calls scipy's highs method) returns exactly $x^* = (0.5, 0.5, 0.5)$, $z_{LP} = 8$ confirming (a)'s feasible point already is the optimum. Gap: $\frac{z^* - z_{LP}}{z^*} = \frac{10 - 8}{10} = 0.2 (20\%)$

Why no int. point can be LP-optimal here. Every int-feasible point costs at least 10 (H1), strictly more than $z_{LP} = 8$; since the LP's feasible region is strictly larger than $\{0,1\}^3$ feasible (it also contains fractional points like (0.5, 0.5, 0.5) which costs less), and the LP optimum is unique at this fractional point, no integral point can achieve the LP optimum. the symmetric fractional point "cheats" each of the three tight covering constraints at once, at half the integral cost of satisfying any two of them outright